

Correlation: What It Is, How to Use It, and Why It Isn't Causation

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1. What correlation measures (and what it does not)

Correlation summarizes the *linear co-movement* between two variables across units, times, or places. In the language of our course: it compares *observed (factual) outcomes* across cases that happen to have different X values.¹ Formally, for random variables X and Y with finite variance,

$$\rho_{XY} = \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X)\text{Var}(Y)}} \quad \text{and} \quad r = \frac{\sum_i (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_i (x_i - \bar{x})^2} \sqrt{\sum_i (y_i - \bar{y})^2}}$$

are the population and sample Pearson correlations, respectively. Key properties:

- $r \in [-1, 1]$; $r = 0$ means no *linear* relationship (nonlinear patterns can exist).
- Unit-free and invariant to positive affine rescaling: $(aX + b, Y) = (X, Y)$ for $a > 0$.
- Sensitive to outliers; rank-based alternatives (Spearman) reduce sensitivity.

2. How to see correlation: plots and a small dataset

Scatter plots are the default visualization; bar/line plots may also be used.² The slides provide a compact data example (firearms crime rate vs. % with gun license by province/territory).³ We reproduce it here for easy plotting in R/Stata/Python:

*This handout follows the Week 10 slides on “Correlation / Confounding” (POLI 110) and adds notation, formulas, and a worked example using the provincial gun-license data. All errors are my own.

¹See the Week 10 slides for the “plug-in” idea: because we cannot see the same unit under different X (the fundamental problem of causal inference), we often look across other cases to substitute the missing counterfactual. Slides pp. 2–4.

²Slides p. 4.

³Slides p. 5.

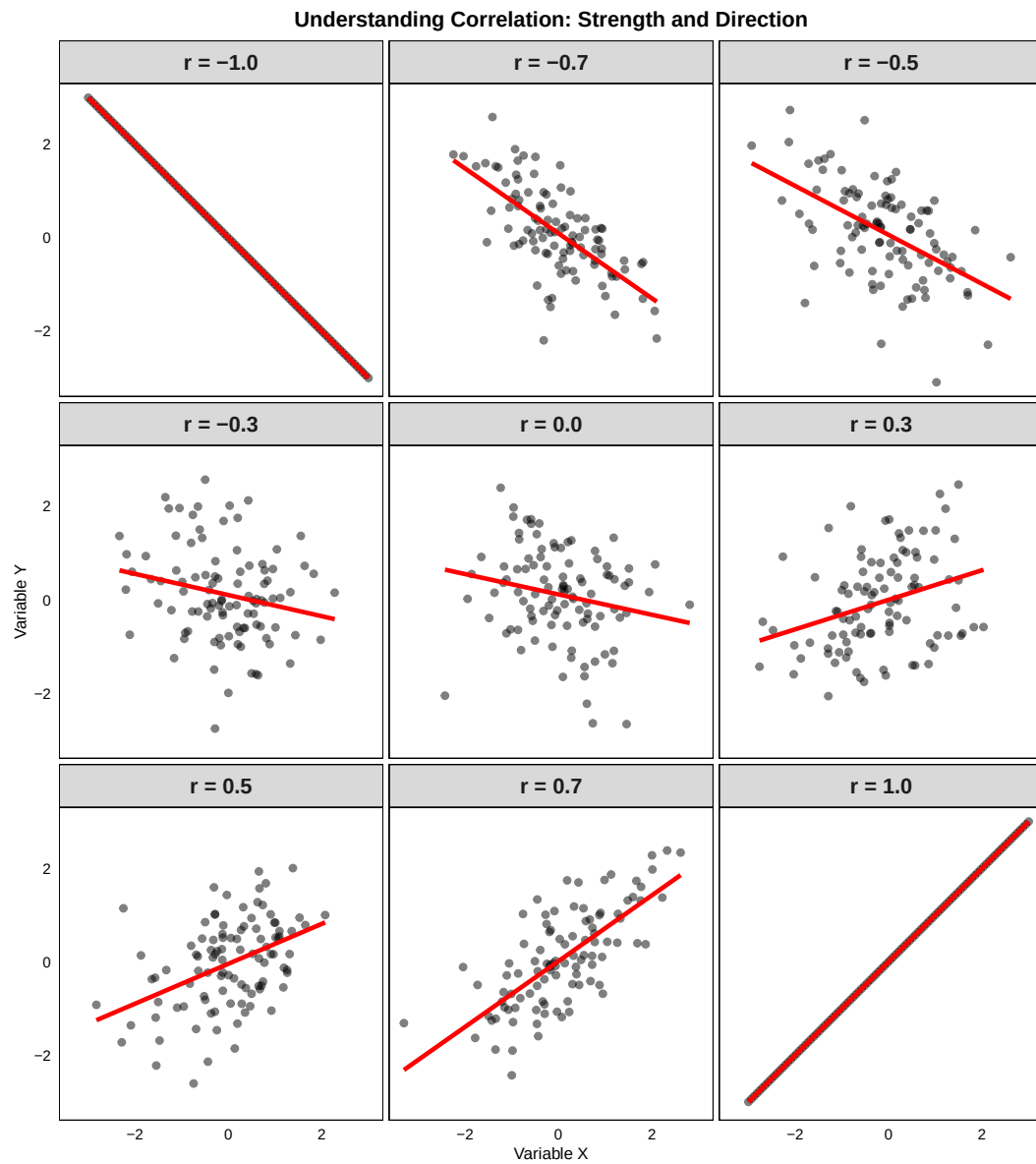


Figure 1: Understanding Correlation: Relationship Strength and Direction.

Province/Territory	% License	Firearms crime rate
Alberta	8	55
B.C.	6	26
Manitoba	7	81
New Brunswick	9	41
Newfoundland & Labrador	14	22
N.W.T.	13	171
Nova Scotia	7	32
Ontario	4	30
P.E.I.	4	26
Quebec	6	25
Saskatchewan	10	127
Yukon	19	71

If you compute the sample correlation with these 12 observations, you obtain approximately $r \approx 0.42$ (a moderate positive linear association), and the simple OLS fit is roughly

$$\widehat{\text{crime}} \approx 19.17 + 4.46 \times (\% \text{ license}).$$

Interpretational warning: this is *not* a causal claim; it is a summary of co-movement across units.

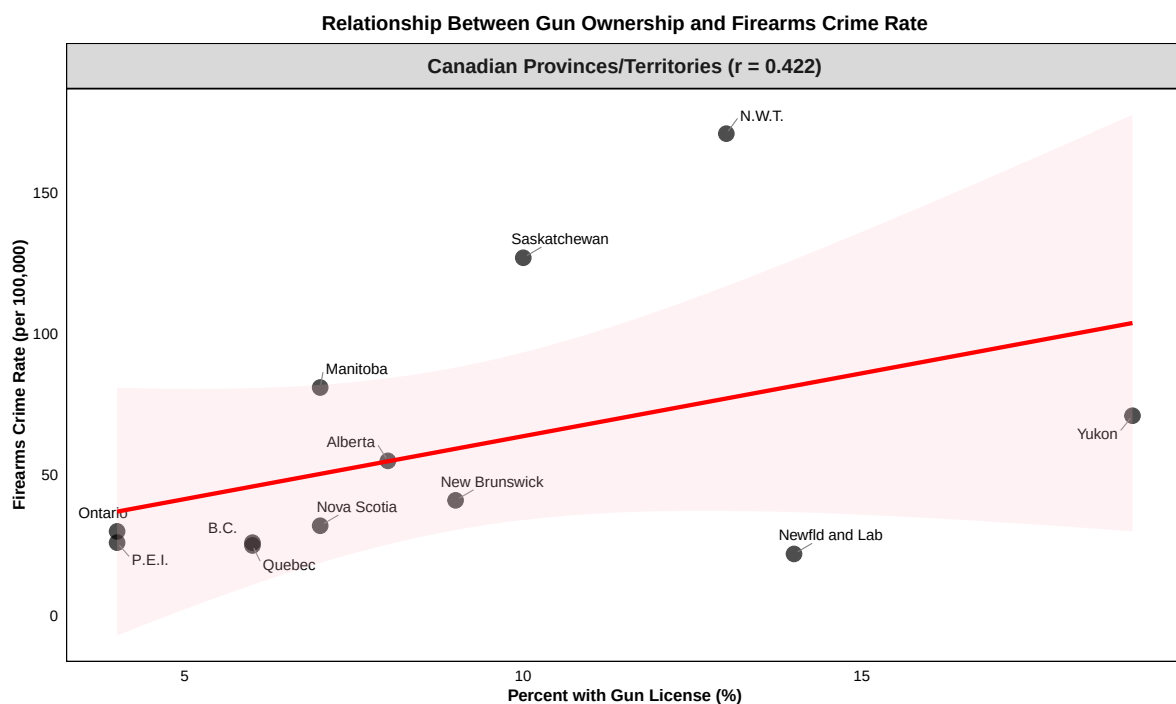


Figure 2: Firearms Crime Rate vs. Gun License Percentage (Canadian Provinces/Territories). Data summarized from Week 10 slide p. 5.

3. Why correlation is not causation

Because we can observe each province only at the license rate it actually has, we cannot see its outcome under higher/lower license rates at the same time. Correlation “plugs in” other provinces to imagine those counterfactuals,⁴ which is informative about *association* but not, by itself, about *causal effects*.

When you see a strong X – Y correlation, map out at least three broad stories (slides p. 10):

- (i) $X \rightarrow Y$ (causation),
- (ii) $Y \rightarrow X$ (reverse causation),
- (iii) A third factor W moves both: $W \rightarrow X$ and $W \rightarrow Y$ (confounding), or pure coincidence.

4. Confounding and the backdoor path

Confounding is a bias in correlation: changes in W shift X and Y in tandem, making X and Y look related even if X has no causal effect on Y .⁵ Causal graphs help detect this: if there is a *backdoor path* $X \leftarrow W \rightarrow Y$, confounding is present (slides pp. 7 and 9).

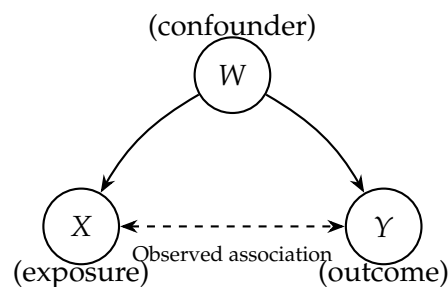


Figure 3: Confounding: W opens a backdoor path between X and Y .

Example 1 (Season, pools, and ice cream). On hot summer days (W), people both go to outdoor pools (X) and buy ice cream (Y), producing a positive X – Y correlation even if swimming itself does not cause ice-cream purchases. The slides also note that indoor pools can flip the sign if $W \rightarrow Y$ is negative.⁶

Example 2 (Policy and outcomes: beware reverse causation). Across jurisdictions, stricter gun laws are often found where gun violence is high. A positive correlation between law strictness and violence could reflect $Y \rightarrow X$ (violence drives legislation), not $X \rightarrow Y$.⁷

⁴Slides pp. 2–3 explain the BC vs. AB illustration of this plug-in logic.

⁵Slides p. 6.

⁶Slides p. 11.

⁷Slides p. 15.

5. Reading correlations in practice: a checklist

When you encounter a correlation (or a trend line in a scatter plot), walk through this short list:

1. **Graph it.** Look for nonlinearity, outliers, and subgroup structure (Anscombe-style pathologies).
2. **List candidate W 's.** Economic development, urbanization, temperature, media exposure, demographics, etc. Draw a quick DAG.
3. **Consider $Y \rightarrow X$.** Policy variables, admissions standards, and enforcement intensity are common outcomes-driven exposures.
4. **Ask: could it be compositional?** Are we comparing provinces, individuals, or time periods with different base rates?
5. **Treat it as a clue, not a verdict.** Use design or modeling to probe causality next.

6. Walkthrough: the provincial example

With the 12-point dataset above, $r \approx 0.42$ and the fitted line is $\hat{y} = 19.17 + 4.46x$. But N.W.T. and Saskatchewan are visually influential; removing or down-weighting such points can move r noticeably. More importantly, even a tight linear pattern would not tell us whether license prevalence *causes* firearms crime. Plausible W 's include rurality/urbanization, policing resources, socioeconomic conditions, and firearms market structure. The slides present several other real-world cross-sectional examples (inequality–corruption; GDP–democracy; immigration share–far-right votes; media consumption–perceived severe weather) precisely to cultivate this habit: see correlation, then reason about W and direction.⁸

7. From correlation to causal claims (very briefly)

Correlation is a starting point. To learn effects, you need identification:

- **Randomization/design:** experiments eliminate backdoor paths by construction.
- **Adjustment:** control for W 's (via regression/matching/weighting) when “no unobserved confounding” is credible.
- **Comparative designs:** difference-in-differences, synthetic controls, instrumental variables, regression discontinuity, etc.

These strategies target *causal* estimands, not just associations.

⁸Slides pp. 12–18.

Appendix A: Formulas and facts

A.1 Covariance and correlation

$$\text{Cov}(X, Y) = \mathbb{E}[(X - \mathbb{E}X)(Y - \mathbb{E}Y)], \quad \text{Var}(X) = \mathbb{E}[(X - \mathbb{E}X)^2].$$

For sample data $\{(x_i, y_i)\}_{i=1}^n$,

$$\widehat{\text{Cov}} = \frac{1}{n-1} \sum_i (x_i - \bar{x})(y_i - \bar{y}), \quad r = \frac{\widehat{\text{Cov}}}{s_X s_Y}.$$

A.2 Invariance

For $a > 0, b, c \in \mathbb{R}$:

$$(aX + b, cY) = \text{sign}(c) (X, Y).$$

Scaling changes slope units but leaves r 's magnitude intact.

A.3 Nonlinearity pitfall

If $Y = g(X)$ with g U-shaped or inverted U, then $r \approx 0$ may coexist with a strong deterministic relationship. Always inspect plots.

References (short)

- F. J. Anscombe (1973), "Graphs in Statistical Analysis," *The American Statistician* 27(1):17–21.
- M. A. Hernán and J. M. Robins (2020), *Causal Inference: What If*. CRC.
- J. Pearl (2009), *Causality* (2nd ed.). Cambridge University Press.